Name: Teacher:	Solutions.	Date:_______

Year 10 OLNA Mathematics
Mini test
Topic – Linear Equations

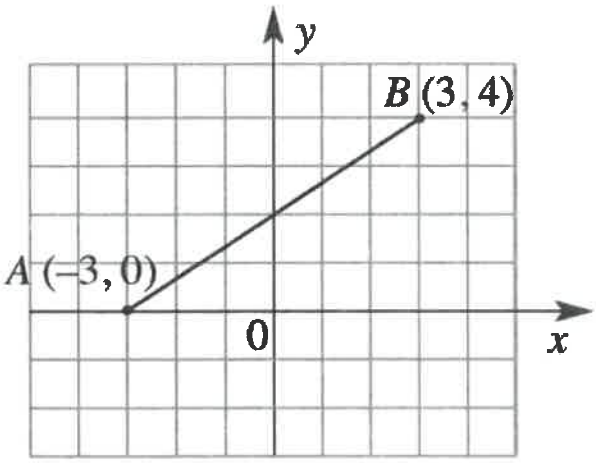
SCORE: /46

Baldivis Secondary College

Full working out MUST be shown to get full marks for each question.

Total Time:	45 minutes	
Weighting:	10 %	
Equipment:	To be provided by the student: Pens and Pencils, Ruler, Scientific calculator, INB.	

1. **[7 marks] Vertical and horizontal distances**
   Consider the following graph



a) What is the **rise** between A and B?

4 ✓

b) What is the **run** between A and B?

6 ✓

c) What is the **gradient** of the line?

✓ $\frac{4}{6} = \frac{2}{3}$ ✓

d) Where does the line cross the y axis?

(0, 2) ✓

e) What is the equation of the line?

$y = \frac{2}{3}x + 2$ ✓ ✓

2. **[16 marks]**
   Complete the following table of values
   $y = 2x + 3$

X	0	1	2	3	4	5	6
Y	3 ✓	5 ✓	7 ✓	9 ✓	11 ✓	13 ✓	15 ✓

$y = -x + 1$

✓ x	0	1	2	3	4	5	6
✓ y	1 ✓	0 ✓	−1 ✓	−2 ✓	−3 ✓	−4 ✓	−5 ✓

3. [10 marks]
Label the following as linear or non-linear. For linear tables, what is the equation of the line?

x	0	1	2	3	4	5	6
Y	3	5	7	9	11	13	17

2 2 2 2 2 2

Linear: Yes / No ✓ **Equation:** $y = 2x + 3$ ✓ ✓

x	-3	-2	-1	0	1	2	3
Y	8	9.25	10.5	11.75	12	12.25	13.5

1.25

Linear: Yes / No ✓ **Equation:** $y = 1.25x + 11.75$ ✓ ✓

x	-3	-2	-1	0	1	2	3
Y	6	4.85	3.7	2.45	1.1	-0.35	-1.9

Linear: Yes / No ✓ **Equation:**

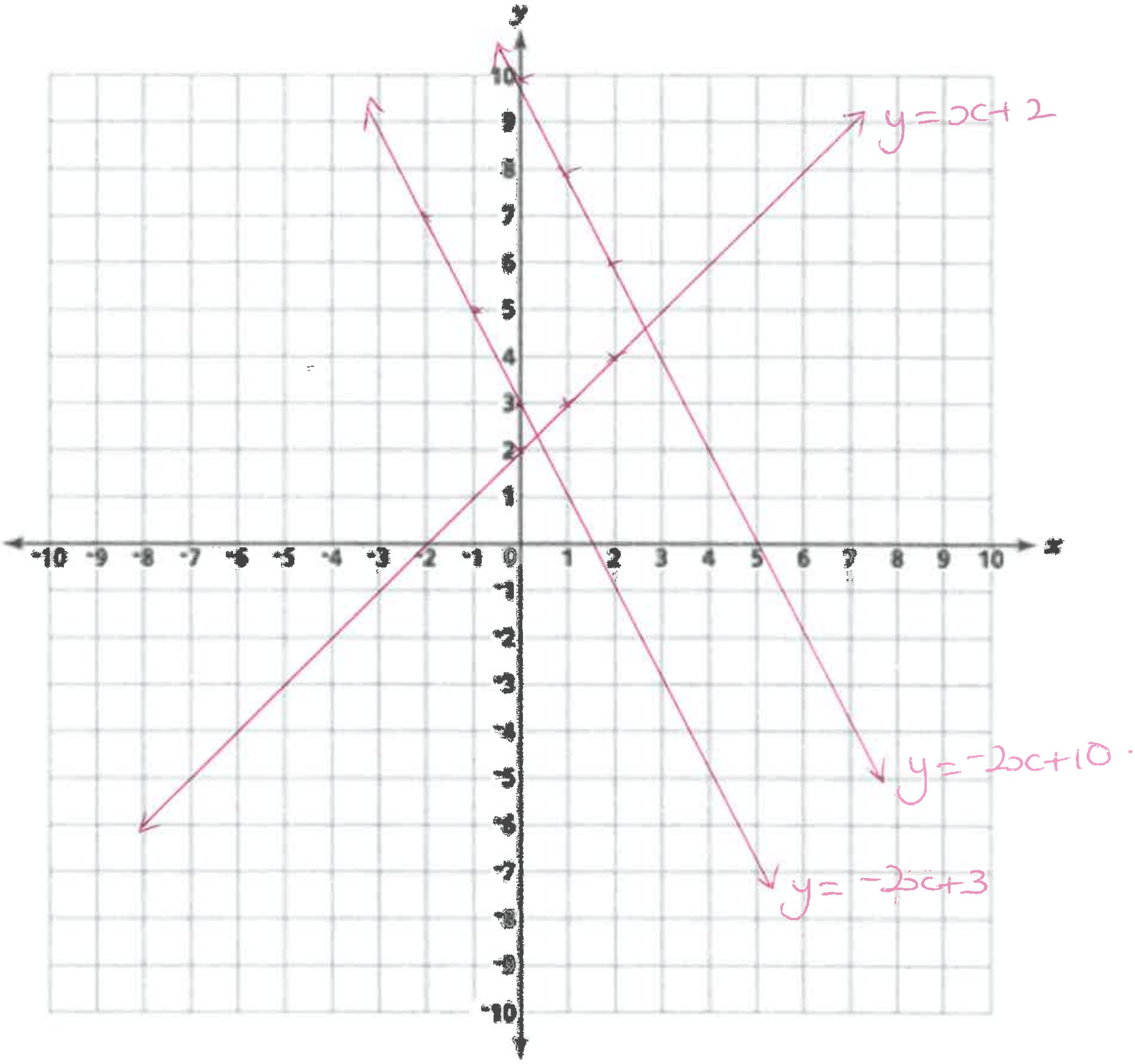
x	-3	-1	2	4	5	6	8
Y	12	10	7	5	4	3	1

Linear: Yes / No ✓ **Equation:** $y = -x + 9$ ✓ ✓

4. **[9 marks]**

Sketch the following graphs on the set of axes (Label your graphs)

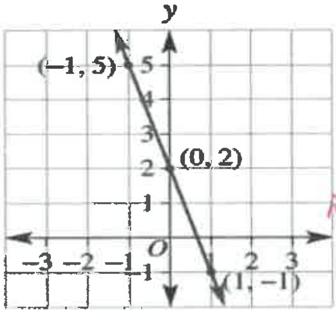
a) $y = x + 2$ b) $y = -2x + 3$ c) $y = -2x + 10$

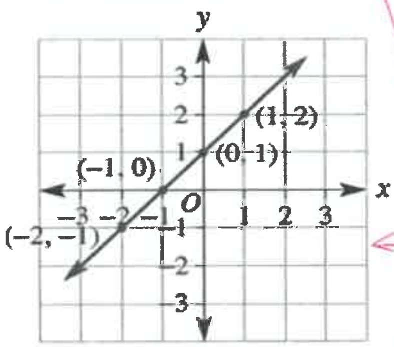


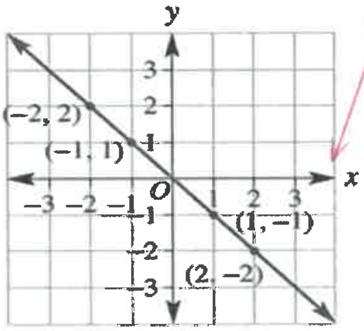
For each line. ✓ gradient ✓ y-intercept ✓ label.

5. [4 marks]

Match the equation to the graph







$x + y = 0$ ✓✓

$y = x + 1$ ✓

$y = -3x + 2$ ✓

End of Test